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(54) **ROTATING ELECTRICAL MACHINE, SET OF SUCH MACHINES, AND ASSOCIATED BOAT AND ROLLING MILL**

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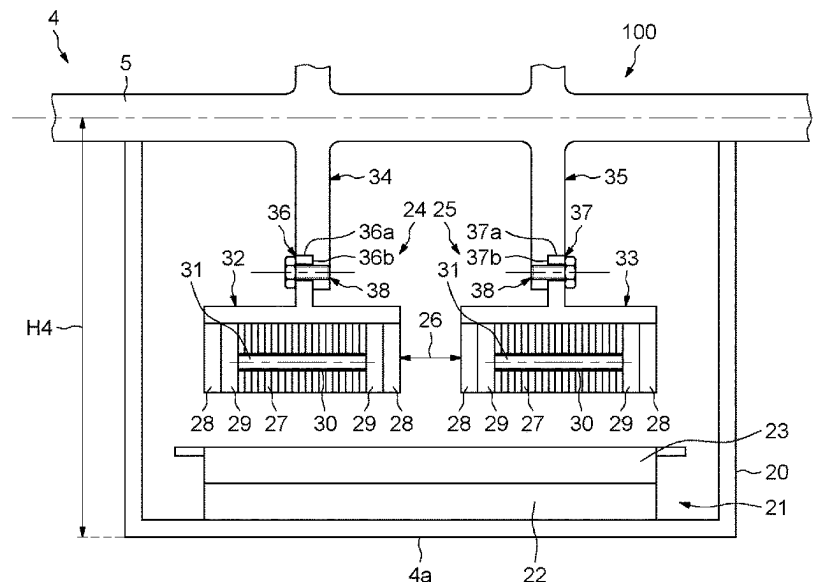
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Primary Examiner — Leda T Pham

(57) **ABSTRACT**

Described herein is a rotating electrical machine, set of such machines, and associated boat and rolling mill. The rotating electrical machine includes a stator, a shaft centered in the stator, a first cylindrical magnetic mass and a second cylindrical magnetic mass, the first cylindrical magnetic mass and the second cylindrical magnetic mass enclosing the shaft and arranged in series on the shaft, the first cylindrical magnetic mass and the second cylindrical magnetic mass being separated by an air gap, the stator including coils, each coil being opposite to the two cylindrical magnetic masses. Each cylindrical magnetic mass includes a stack of compacted laminated magnetic sheets, first fastening means configured to fix the first cylindrical magnetic mass and the shaft, and second fastening means configured to fix the second cylindrical magnetic mass and the shaft.

14 Claims, 8 Drawing Sheets



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FIG.1

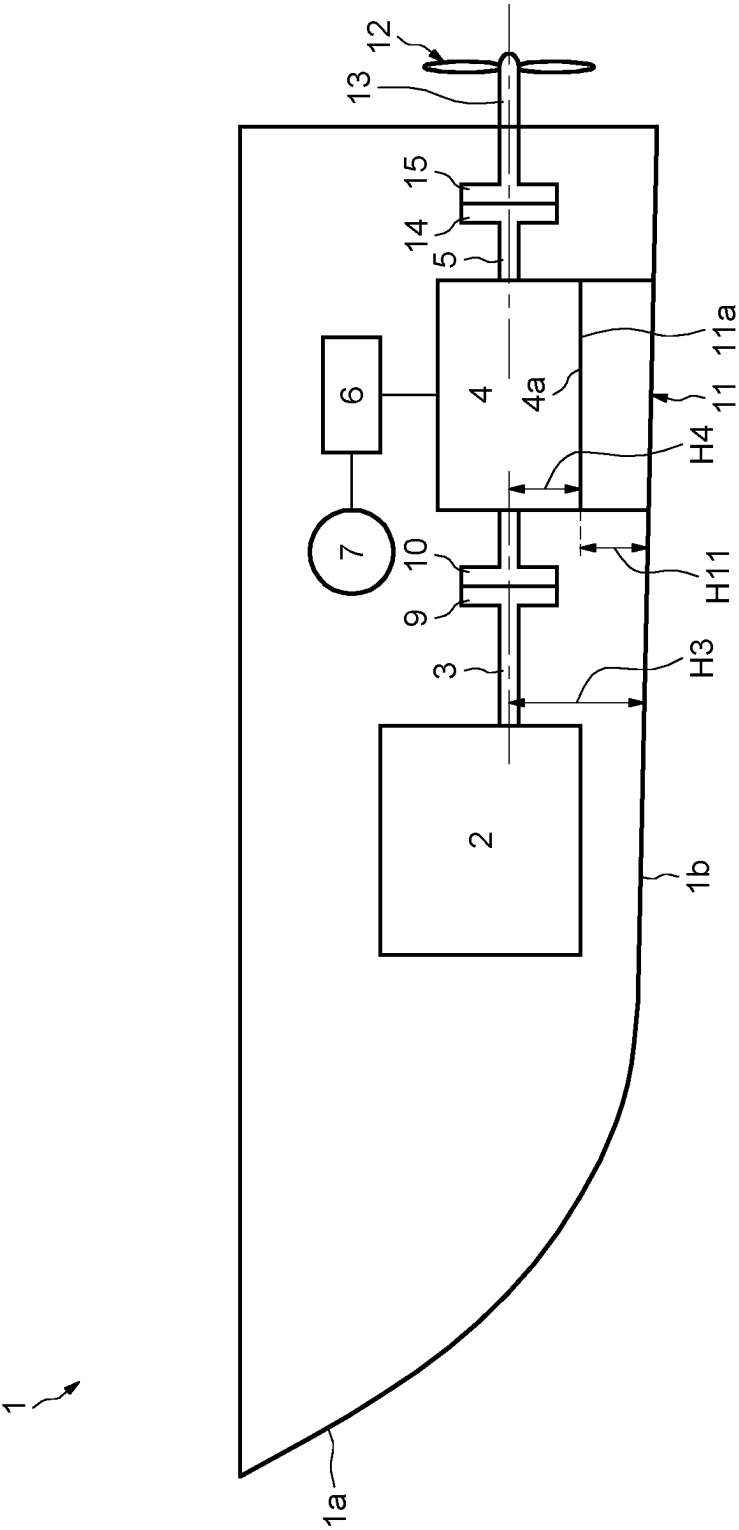


FIG.2

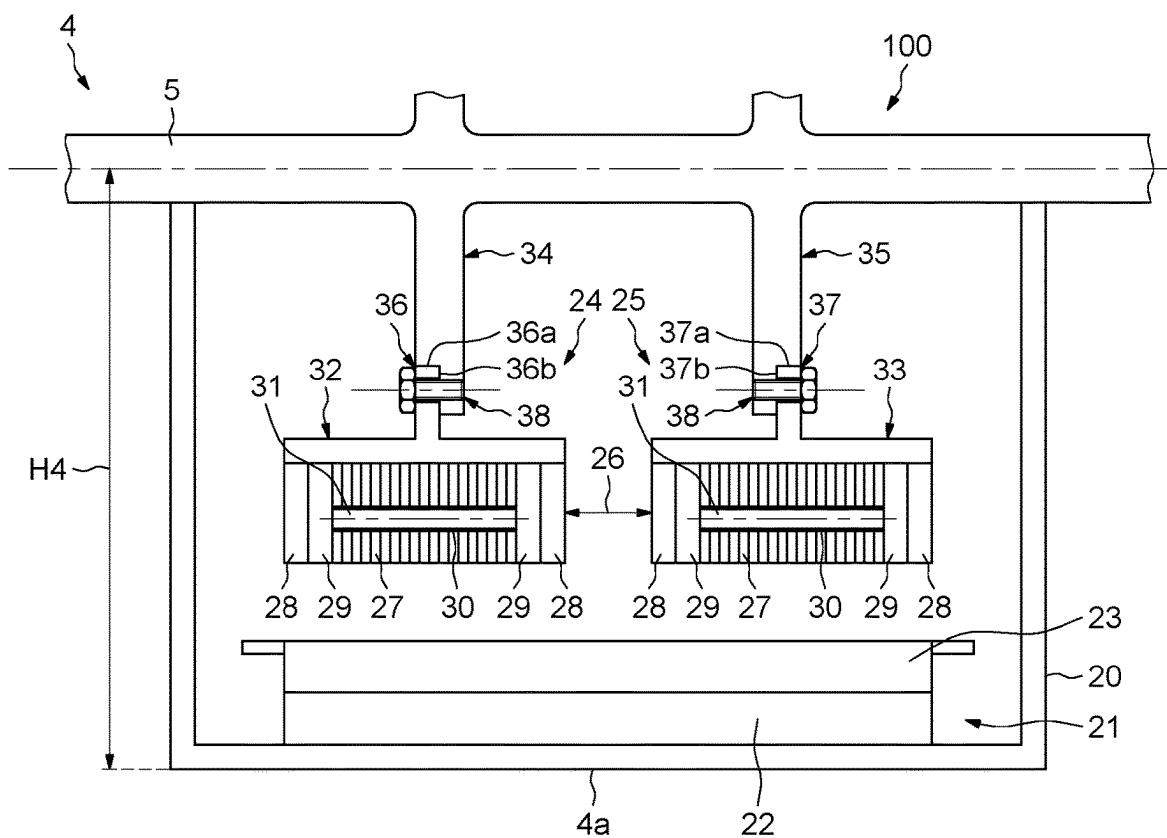


FIG.3

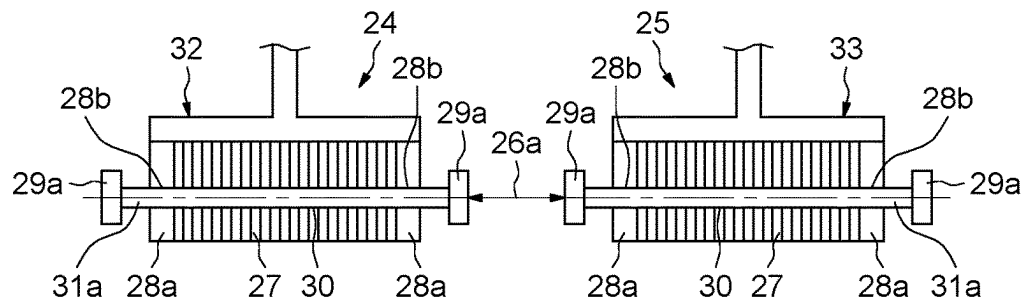


FIG.4

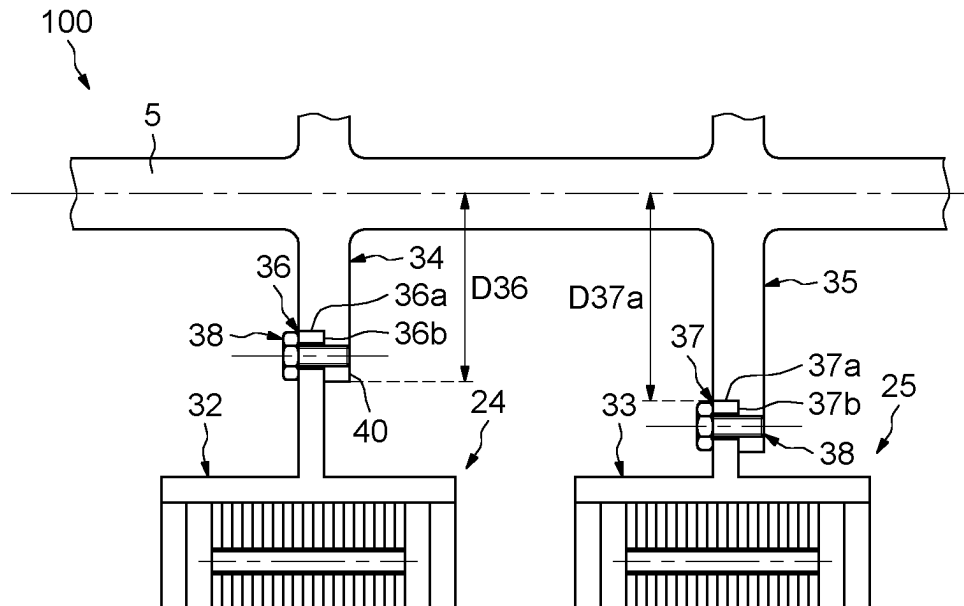


FIG.5

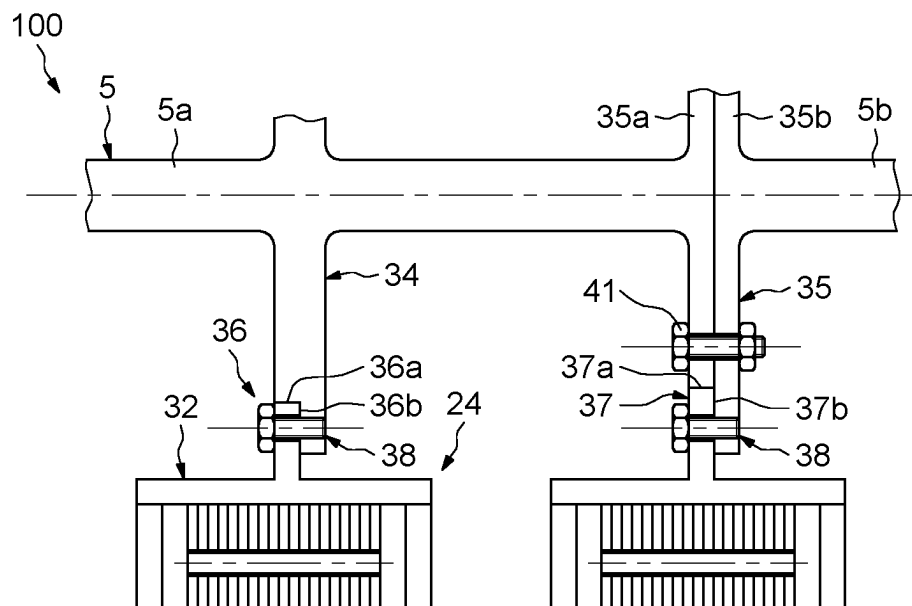


FIG.6

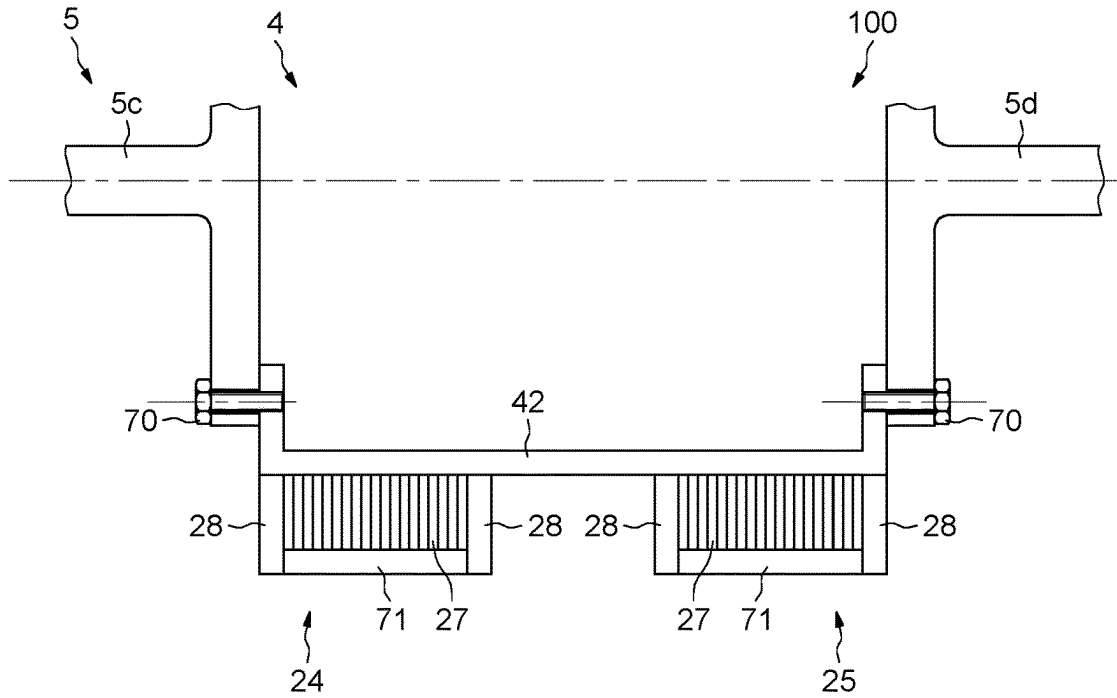


FIG.7

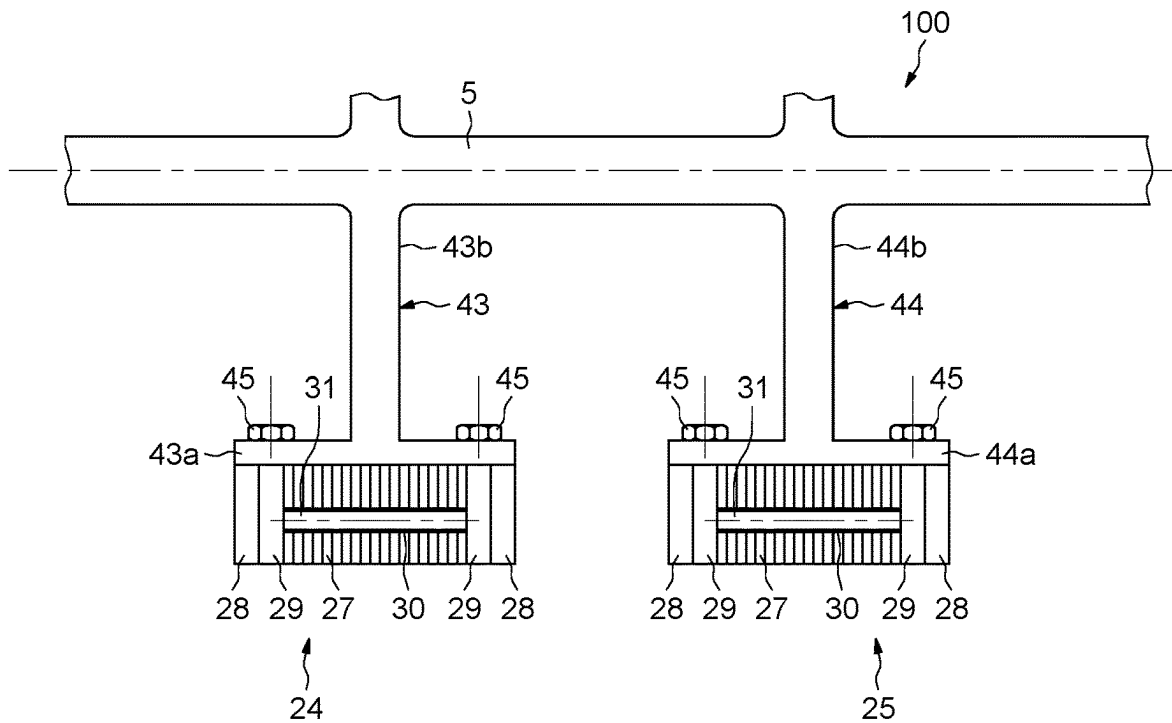


FIG.8

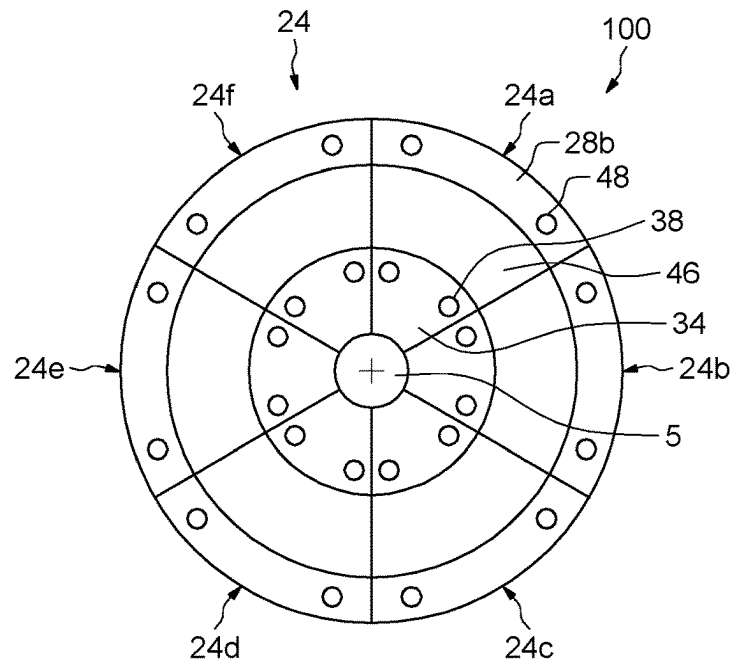


FIG.9

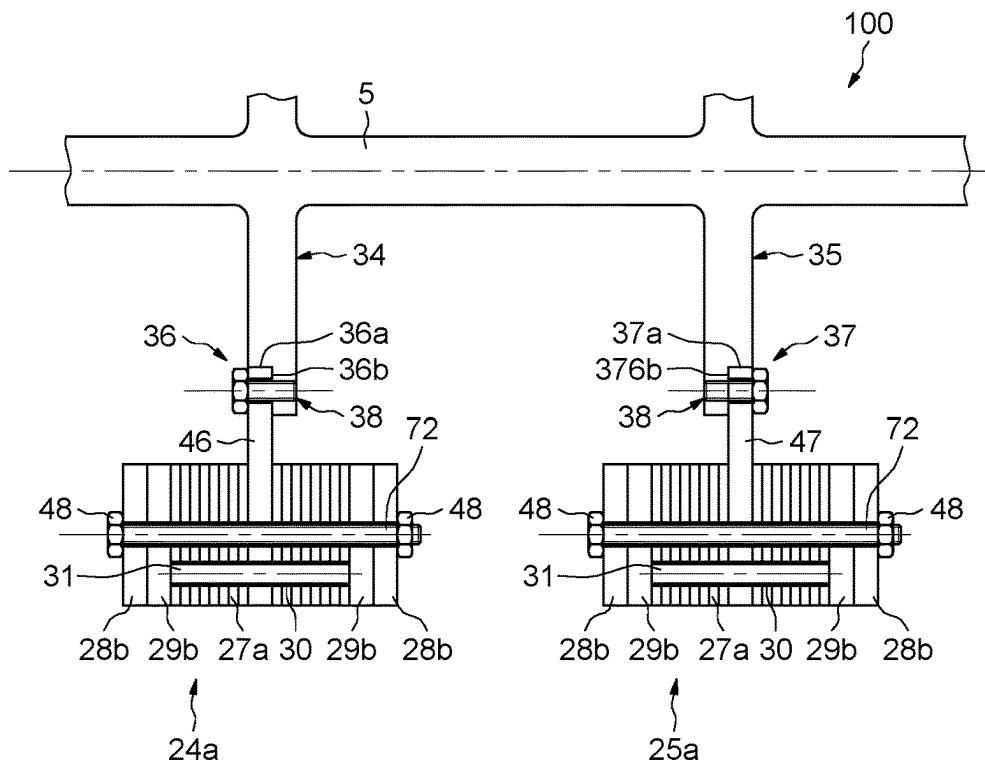


FIG.10

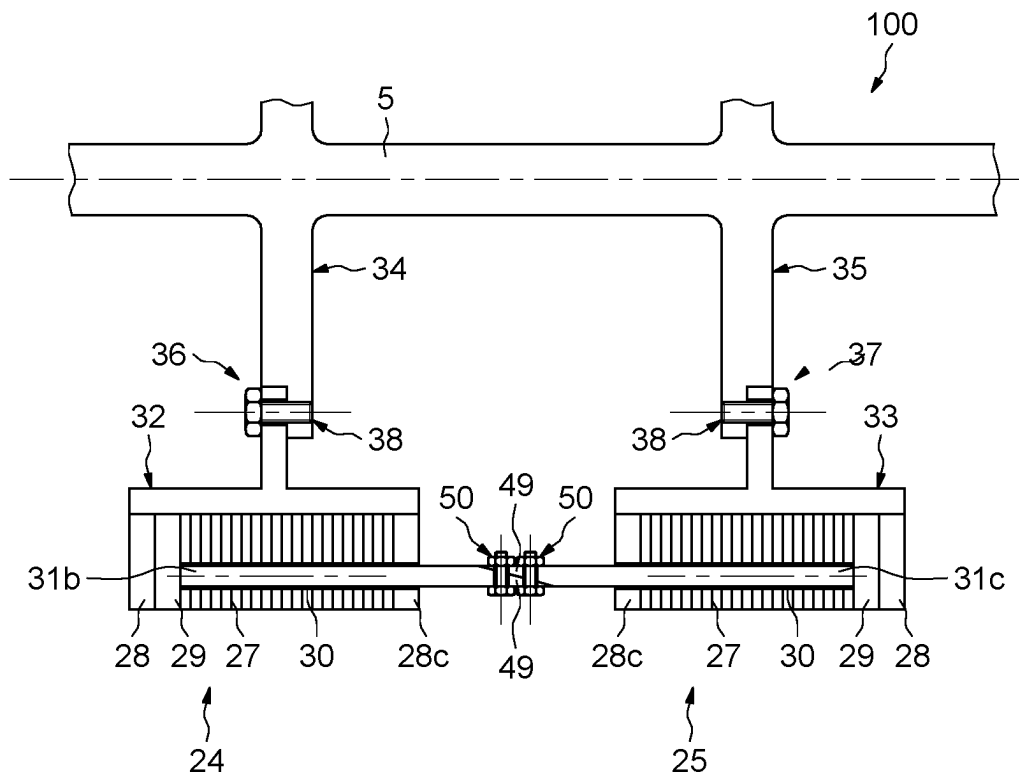


FIG.11

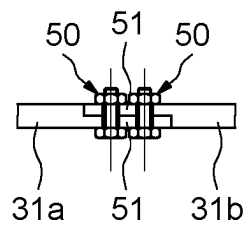


FIG.12

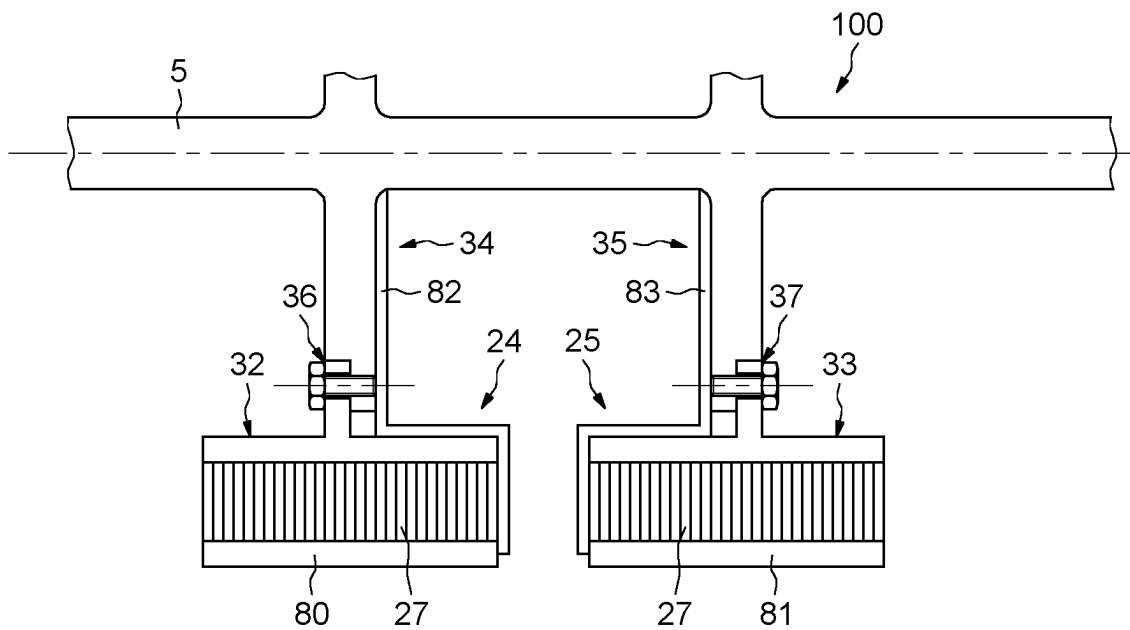


FIG.13

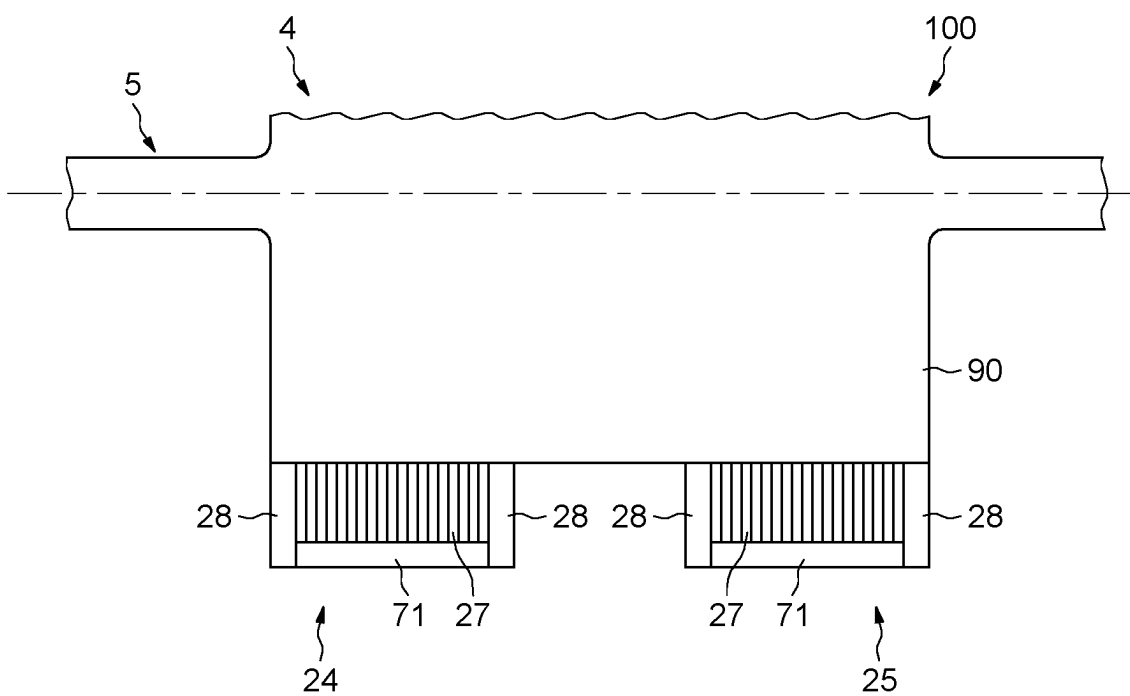
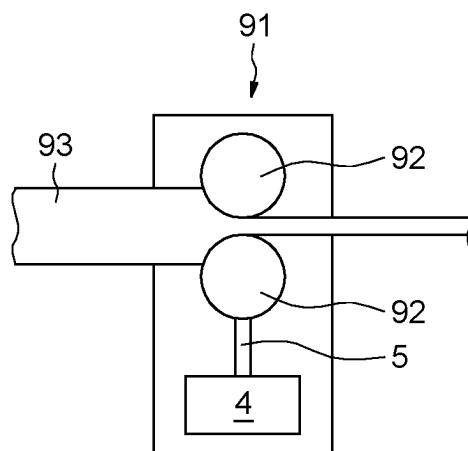


FIG. 14



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ROTATING ELECTRICAL MACHINE, SET OF SUCH MACHINES, AND ASSOCIATED BOAT AND ROLLING MILL

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a national stage entry of European Application No. EP22306869.3, filed Dec. 13, 2022, which is incorporated herein by reference.

FIELD OF THE DISCLOSURE

The present disclosure concerns rotating electrical machines and relates more particularly to a rotating electrical machine having a stator and a shaft comprising two cylindrical magnetic masses in tandem, a boat comprising such a rotating electrical machine, and, a rolling mill comprising such an electrical rotating machine.

BACKGROUND OF THE DISCLOSURE

In this description, “boat” shall mean any type of motorized floating vessel or vehicle designed to be sailed.

A boat may comprise a diesel engine comprising a driving shaft driving a propeller of the boat and an electrical rotating machine used between the diesel engine and the propeller to add propulsive torque to the shaft in a motor mode or to add electrical power to the network in a generator mode.

In another embodiment, the boat comprises an electrical rotating machine driving a propeller of the boat, the boat having no diesel engine.

The electrical machine may reach up a power to 6 MW at a rotation speed varying from 50 rpm to 250 rpm according to the type of boat.

Generally, the rotating electrical machine comprises a stator and a rotor comprising a rotor shaft comprising a shaft collar and a magnetic mass fixed on the shaft collar generally with bolts.

The rotor shaft is connected at each of its ends to the driving shaft and between the propeller and the diesel engine or is connected at one of its ends to the propeller if the boat does not comprise a diesel engine.

The rotating electrical machine is generally an induction machine with a wounded rotor or a squirrel cage rotor, or a synchronous machine made of either a permanent magnet rotor or a wounded rotor.

As the required torque generated by the said machine in the motor mode or the electrical power generated by the said machine in the generator mode is still increasing, it is known to increase the diameter of the magnetic mass and of the stator to meet such requirements.

As the electrical machine may be implemented in the boat, the space reserved for each machine is limited.

Generally, the boat architect determines a maximal distance between an electrical machine bracket on which lies the electrical machine and the center of the driving shaft.

The maximal distance imposes the maximal diameter of the magnetic mass.

As the maximal diameter of the magnetic mass is limited, the maximal torque and the electrical power generated by the electrical machine is limited and may not meet the required generated torque and the required generated electrical power.

Further, as the forces applied on the bolts connecting the rotor magnetic mass to the rotor shaft are proportional to the inertia of the magnetic mass and to the rotating speed of the

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rotor, when the shaft is decelerating, the rotor may reach a deceleration up to $40 \text{ rad}\cdot\text{s}^{-2}$ generating huge torques on the bolts susceptible to deteriorate the bolts.

As the inertia of such a rotor is significant, the rotating inertia of other elements connected to the driving shaft, such as the propeller, may generate torsional resonance on the rotor shaft and on the driving shaft.

The torsional resonance may reach torsional acceleration up to $40 \text{ rad}\cdot\text{s}^{-2}$ generating huge torques on the bolts connecting the magnetic mass to the rotor shaft.

Such a rotating electrical machine may also be implemented in a rolling mill to drive rolls to roll steel strands.

The rotating electrical machine implemented in a rolling mill may reach up a power to 30 MW at a rotation speed varying from 100 rpm to 200 rpm depending of the type of rolling mill.

As the inertia of such a rotor is significant, the other main inertia connected to the shaft driving the rolls may generate torsional resonance on the rotor shaft of the said rotating electrical machine.

The torsional resonance may reach torsional acceleration up to $40 \text{ rad}\cdot\text{s}^{-2}$ generating huge torques on the bolts connecting the magnetic mass to the rotor shaft.

Further, hen steel strands are inserted between the rolls, the rotor may reach a deceleration up to $100 \text{ rad}\cdot\text{s}^{-2}$.

The document U.S. Pat. No. 7,797,815 discloses a permanent magnet electrical machine comprising a shaft assembly made of a first shaft comprising permanent magnets forming a first rotor and a second shaft comprising permanent magnets forming a second rotor.

As the permanent magnets are fixed on the first and second shaft to from the first and second rotors, the rotors are not removable.

Further, magnetic fields generate current losses (iron losses) in the first and second shafts deteriorating the efficiency of the permanent magnet electrical machine.

Transporting, manufacturing, assembling or repairing a permanent magnet rotor can be very complex and requires non-magnetic tools.

The permanent magnets may be replaced by coils to form a synchronous machine.

However, the first and second rotors are not demountable.

Further, the magnetic fields generate current losses (iron losses) in the first and second shafts.

It is therefore proposed to remedy in whole or in part the disadvantages related to the electrical machines comprising magnetic masses fixed on shaft collars of rotor shafts according to the prior art.

SUMMARY OF THE DISCLOSURE

The present disclosure discloses a rotating electrical machine comprising a stator, a shaft centered in the stator, a first cylindrical magnetic mass and a second cylindrical magnetic mass, the first cylindrical magnetic mass and the second cylindrical magnetic mass enclosing the shaft and arranged in series on the shaft, the first cylindrical magnetic mass and the second cylindrical magnetic mass being separated by an air gap, the stator including coils, each coil being opposite to the two cylindrical magnetic masses.

Each cylindrical magnetic mass comprises a stack of compacted laminated magnetic sheets, first fastening means configured to fix the first cylindrical magnetic mass and the shaft, and second fastening means configured to fix the second cylindrical magnetic mass and the shaft.

Advantageously, each cylindrical magnetic mass comprises a set of coils so that the first cylindrical magnetic mass

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forms a first wounded rotor and the second cylindrical magnetic mass forms a second wounded rotor.

Preferably, each cylindrical magnetic mass comprises a set of coils, each coil of the set of coils of the first cylindrical magnetic mass being connected to a different coil of the set of coils of the second cylindrical magnetic mass so that the first and second cylindrical magnetic masses form a wounded rotor.

Advantageously, each cylindrical magnetic mass comprises conductive bars and two short circuit disks or two short circuit rings to form a squirrel cage, the first cylindrical magnetic mass forming a first squirrel cage rotor and the second cylindrical magnetic mass forming a second squirrel cage rotor.

Preferably, the first cylindrical magnetic mass comprises a first set of conductive bars and a first short circuit disk and the second cylindrical magnetic mass comprises a second set of conductive bars and a second short circuit disk, each conductive bar of the first set being connected to the first short circuit disk and to a different conductive bar of the second set, the conductive bars of the second set being connected to the second short circuit disk, the first and second sets and the first and second short circuit disks forming a squirrel cage so that the first and second cylindrical magnetic masses form a squirrel cage rotor.

Advantageously the end of each conductive bar of the first set and the second set in contact with another conductive bar comprises a slanted slide.

Preferably, the diameter of the first cylindrical mass is equal to the diameter of the second cylindrical mass, the rotating electrical machine further comprises:

- a first full rim including a first central circular aperture, the first magnetic mass being fastened on the periphery of the first full rim,

- a second full rim including a second central circular aperture, the second magnetic mass being fastened on the periphery of the second full rim,

- a first shaft collar and a second shaft collar on the shaft, wherein:

- the first fastening means comprises a first shoulder on the periphery of the first shaft collar and first fixing means, the first full rim is into abutment against the first shoulder and is fixed to the first shaft collar by the first fixing means,

- the second full rim is into abutment against the second shoulder and is fixed to the second shaft collar by the second fixing means,

- the first shoulder comprises a first cylindrical surface in contact with the circumferential surface of the first central circular aperture and a first annular surface perpendicular to the first cylindrical surface in contact with the first full rim, and

- the second shoulder comprises a second cylindrical surface in contact with the circumferential surface of the second central circular aperture and a second annular surface perpendicular to the second cylindrical surface and in contact with the second rim.

Advantageously, the diameter of the first shaft collar is smaller than the diameter of the second cylindrical surface, and the first and second shoulders are oriented on the first and second shaft collars so that the second annular surface is facing the surface of the first shoulder opposite to the first annular surface of the first shoulder.

Preferably, the shaft comprises two half shafts, and wherein one shaft collar of the first and second shaft collars comprises two parts, a first half shaft comprising a first part and the second half shaft comprising the second part, the

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first part comprising the annular surface or the cylindrical surface of the shoulder of the said one shaft collar and the second part comprising the cylindrical surface or the annular surface of the shoulder of the said one shaft collar, the first and second half shafts being fixed together with second removable fastening means to form the shoulder of the said shaft collar, the first half shaft or the second half shaft comprising the other shaft collar of the first and second shaft collars.

Advantageously, the diameter of the first cylindrical mass is equal to the diameter of the second cylindrical mass, the rotating electrical machine further comprising a hollow cylindrical support part, the shaft comprising a first half shaft and a second half shaft, the cylindrical support part being inserted between the first half shaft and the second half shaft, and the first cylindrical magnetic mass and the second cylindrical magnetic mass being fixed on the periphery of the cylindrical support part, the first and second fastening means comprising the hollow cylindrical support part.

Preferably, the rotating electrical machine further comprises a third full rim and a fourth full rim enclosing the shaft, the first cylindrical magnetic mass being fixed on the periphery of the third full rim with the first fastening means, and the second cylindrical magnetic mass being fixed on the periphery of the fourth full rim with the second fastening means.

Advantageously, at least one magnetic mass of the first cylindrical magnetic mass and the second cylindrical magnetic mass is segmented in segments, each segment comprising a stack of segmented laminated magnetic sheets and a segmented support part separating the stack in two sets of segmented laminated magnetic sheets, the fastening means associated to the one magnetic mass fixing each support part on the shaft.

Another object of the invention relates to a set of rotating electrical machines comprising at least a first rotating electrical machine as defined above and a second rotating electrical machine as defined above, the shaft of the first rotating electrical machine being connected to the shaft of the second rotating electrical machine.

An object of the invention relates to a boat comprising at least one rotating electrical machine as defined above.

Another object of the invention relates to a rolling mill comprising at least one rotating electrical as defined above.

BRIEF DESCRIPTION OF THE DRAWINGS

Other characteristics and advantages of the invention will emerge on reading the following description of embodiments of the invention, provided solely by way of non-limiting examples and with reference to the drawings in which:

FIG. 1 illustrates an example of a boat according to the invention,

FIG. 2 illustrates a partial longitudinal cross section of an exemplary embodiment of an electrical machine according to the invention,

FIG. 3 illustrates a variant of magnetic masses according to the invention,

FIG. 4 illustrates a partial longitudinal cross section of a second example of a moving set according to the invention,

FIG. 5 illustrates a partial longitudinal cross section of a third example of the moving set according to the invention,

FIG. 6 illustrates a partial longitudinal cross section of a fourth example of the moving set according to the invention,

FIG. 7 illustrates a partial longitudinal cross section of a fifth example of the moving set according to the invention,

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FIGS. 8 and 9 illustrate a partial longitudinal cross section of a sixth example of the moving set according to the invention,

FIG. 10 illustrates a partial longitudinal cross section of a seventh example of the moving set according to the invention,

FIG. 11 illustrates another embodiment of the ends of conductive bars according to the invention,

FIG. 12 illustrates a partial longitudinal cross section of an eighth example of the moving set according to the invention,

FIG. 13 illustrates a partial longitudinal cross section of a ninth example of the moving set according to the invention, and

FIG. 14 illustrates an example of a rolling mill according to the invention.

DETAILED DESCRIPTION

Throughout the application, the term “invention” has been used interchangeably with the term “disclosure”.

FIG. 1 illustrates an example of a boat 1 comprising a hull 1a having a base line 1b.

The boat further comprises a diesel engine 2 including an engine shaft 3, a rotating electrical machine 4 including a shaft 5, a reversible power converter 6 connected to the rotating electrical machine 4, and a power supply network 7 connected to the reversible power converter 6.

One end of the engine shaft 3 is connected to a first end of the shaft 5 for example with two coupling flanges 9, 10. One coupling flange 9 is at the said end of the engine shaft 3 and the other coupling flange 10 is at the first end of the shaft 5.

The distance between the base line 1b and the center of the engine shaft 3 is denoted H3 and may be determined by the boat architect according for example to the dimensions of the diesel engine 2.

The boat 1 further comprises a bracket 11 laying on the bottom of the hull 1a and a propeller 12.

The rotating electrical machine 4 is disposed on an upper surface 11a of the bracket 11 so that a base surface 4a of the rotating electrical machine 4 is in contact with the upper surface 11a of the bracket 11.

The upper surface 11a of the bracket 11 is opposed to the surface of the bracket 11 in contact with the bottom of the hull 1a.

The distance between the center of the shaft 5 of the machine 4 and the upper surface 11a of the bracket 11 is denoted H4.

The height between the base line 1b and the upper surface 11a of the bracket 11 at the closest end of the rotating electrical machine 4 to the diesel engine 2 is denoted H11.

The propeller 12 comprises a propeller shaft 13 connected to the second end of the shaft 5 for example with two coupling flanges 14, 15. One coupling flange 14 is at the second end of the shaft 5 of the machine 5 and the other coupling flange 15 is at the end of the propeller shaft 12.

The engine shaft 3 and the shaft 5 of the machine 4 form a driving shaft driving the propeller 12 to propel the boat 1.

The machine 4 may operate in a motor mode or in a generator mode.

In the motor mode, the machine 4 supplied by the power supply network 7 through the reversible power converter 6 generates a torque on the shaft 5 to assist the diesel engine 2.

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In the generator mode, the shaft 5 of the machine 4 is driven by the engine shaft 3 so that the machine 4 generates electrical energy supplying the reversible power converter 6.

The power converter 6 supplies for example the power supply network 7 with the electrical energy power generated by the machine 4 in generator mode.

The boat 1 may comprise more than one network 7.

The reversible power converter 6 may be made from semiconductors, for example diodes, thyristors or piloted interrupters.

The diesel engine 2 may be replaced by a driving system generating a torque to drive the engine shaft 3, for example a propulsion electrical machine or a gas motor.

The shaft 5 and one or both of the engine shaft 3 and the propeller shaft 12 may be made of one shaft.

As the distances H3 and the height H11 are imposed, the distance H4 between the center of the shaft 5 of the machine 4 and the upper surface 11a of the bracket 11 is maximal equal to the distance H3 between the base line 1b and the center of the engine shaft 3 minus the height H11.

The torque generated by the machine 4 in motor mode is proportional to the magnetic fluxes generated by a stator and a rotor of the machine 4, in other words the torque generated by the machine 4 in motor mode is proportional to the circumferential surface of the rotor in opposite to the circumferential surface of the stator so that the torque generated by the machine 4 depends on the diameter of the stator and the diameter of the rotor.

As the distance H4 is imposed, the diameter of the stator and the diameter of the rotor are imposed.

FIG. 2 illustrates a partial longitudinal cross section of an exemplary embodiment of the machine 4.

The machine 4 comprises a casing 20 having the base surface 4a.

The distance H4 between the center of the shaft 5 and the base surface 4a is maximal equal to the distance H3 minus the height H11.

The casing 20 encloses a stator 21 comprising a stator frame 22 and a stator magnetic mass 23, both being made for example of compacted magnetic laminated sheets.

The shaft 5 is centered in the stator 21.

The stator magnetic mass 23 comprises coils connected to the power converter 6.

The casing 20 further encloses a first embodiment of a moving set 100 comprising the shaft 5, a first cylindrical magnetic mass 24, and a second cylindrical magnetic mass 25.

The first cylindrical magnetic mass 24 and the second cylindrical magnetic mass 25 enclose the shaft 5 and are arranged in series on the shaft 5 so that the first cylindrical magnetic mass 24 and the second cylindrical magnetic mass 25 are separated by an axial air gap 26.

The first and second cylindrical magnetic masses 24, 25 are arranged in tandem on the shaft 5 and may have an identical diameter so that the radial air gap between the first and second cylindrical magnetic masses 24, 25 and the stator 21 is equal.

Each coil of the stator 21 extends along a direction longitudinal direction parallel to the shaft 5 so that each coil of the stator 21 is opposite to the first and second cylindrical magnetic masses 24, 25.

Each cylindrical magnetic mass 24, 25 comprises laminated magnetic sheets 27.

The first cylindrical magnetic mass 24 and the shaft 5 are fixed with first fastening means and the second magnetic mass 25 are fixed with second fastening means.

The first fastening means and the second fastening means may be identical or different.

Each magnetic mass **24**, **25** comprises two compaction plates **28** connected together with ties rods (not represented) and tightening the magnetic laminated sheets to maintain the stacked magnetic laminated sheets compacted.

Each magnetic mass **24**, **25** further comprises two short circuit disks **29**, each short circuit disk **29** being inserted between one different compaction plate **28** and one end of the stack of magnetic laminated sheets **27**.

In variant, the short circuit disks **29** may be replaced by short circuit rings.

Each magnetic mass **24**, **25** further comprises housings **30** extending along the longitudinal direction and regularly disposed on a diameter of the said magnetic mass, each housing **30** lodging a conductive bar **31**.

The short circuit disks **29** and a first set of conductive bars **31** of the first magnetic mass **24** form a first squirrel cage so that the first magnetic mass **24** and the shaft **5** form a first squirrel cage rotor.

Similarly, the short circuit disks **29** and a second set of conductive bars **31** of the second magnetic mass **24** form a second squirrel cage so that the shaft **5** and the second magnetic mass **25** form a second squirrel cage rotor.

In this exemplary embodiment, the machine **4** is an asynchronous machine comprising two squirrel cage rotors.

In variant, the squirrel cage of the first and second magnetic masses **24**, **25** may be replaced with a first and a second set of coils so that the machine **4** is an asynchronous machine comprising two wounded rotors or so that the machine **4** is a synchronous machine comprising two wounded rotors, for example a salient pole synchronous machine.

In another variant, the squirrel cage of the first and second magnetic masses **24**, **25** may be replaced with permanent magnets so that machine **4** is a permanent magnet machine comprising two permanent magnet rotors.

The machine **4** may further comprise a first full rim **32** including a first central circular aperture and a second full rim **33** including a second central circular aperture.

The first full rim **32** and the second full rim **33** may be identical.

The first magnetic mass **24** is fastened on the periphery of the first full rim **32** and the second magnetic mass **25** is fastened on the periphery of the second full rim **33**.

The shaft **5** may further comprise a first shaft collar **34** and a second shaft collar **35**.

The first shaft collar **34** and the second shaft collar **35** may be identical.

The first shaft collar **34** may comprise a first shoulder **36** on its periphery, the first shoulder **36** comprising a first cylindrical surface **36a** and a first annular surface **36b** perpendicular to the first cylindrical surface **36a**.

Similarly, the second shaft collar **35** may comprise a second shoulder **37** on its periphery, the second shoulder **37** comprising a second cylindrical surface **37a** and a second annular surface **37b** perpendicular to the second cylindrical surface **37a**.

The first full rim **32** is into abutment with the first shoulder **36** so that the first cylindrical surface **36a** is in contact with the circumferential surface of the first central circular aperture and the first annular surface **36b** is in contact with the first full rim **32**.

The first full rim **32** is fixed to the first shaft collar **34** with first fixing means.

The first fastening means comprise the first shoulder **36** and the first fixing means.

The second full rim **33** is into abutment with the second shoulder **37** so that the second cylindrical surface **37a** is in contact with the circumferential surface of the second central circular aperture and the second annular surface **37b** is in contact with the second full rim **33**.

The second full rim **33** is fixed to the second shaft collar **35** with second fixing means.

The second fastening means comprise the second shoulder **37** and the second fixing means.

The first and second fixing means may be identical or different.

The first and second fixing means may comprise removable fixing means, for example screws **38** passing through the first and second full rim **32**, **33** and being engaged in the first and second shaft collars **34**, **35** so that the first and second fastening means are removable first and second fastening means.

In variant, the removable fixing means may comprise bolts.

The use of removable fixing means permits to easily dismount the first and second magnetic masses **24**, **25** to facilitate transporting, manufacturing, assembling or repairing of the machine **4**.

In variant, the first and second fixing means may comprise fixed fastening means so that the first and second fastening means are fixed first and second fastening means, the full rim being not dismantlable without deteriorating a rim and the associated shaft collar.

Fixed fastening means comprise for example solder between the said shaft collar and the said full rim.

The use of laminated magnetic sheets permits to reduce the current losses in the first and second rotors and the eddy currents.

Further, as the distance **H4** is imposed, the machine **4** comprising two magnetic masses permits to increase the surface of magnetic masses in regard with the coils of the stator **21** to increase the torque generated by the machine **4** in motor mode or the electrical energy generated by the machine **4** in generator mode compared to a machine comprising a housing including a stator and only one magnetic mass known from the prior art and having a distance between the center of the rotor shaft and the external surface of the housing laying on a bracket equal to the distance **H4**.

The arrangement of two magnetic masses **24**, **25** on the shaft **5** further permits to distribute magnetic forces applied on the shaft **5** when the coils of the stator **21** are supplied to limit the deformation of the shaft **5** suppressing intermediate bearings generally arranged in machines known in the prior art.

The diameter of the magnetic masses **24**, **25** is reduced compared to the diameter of a rotor magnetic mass of a machine known from the prior art generating the same torque or electrical energy as the machine **4** so that the inertia of the magnetic masses **24**, **25** is reduced compared to the inertia of the rotor magnetic mass of a machine known from the prior art. The torque applied and the first and second fastening means during deceleration of the shaft **5** is reduced, and the occurrence of torsional resonance on the shaft **5** and on the driving shaft is reduced allowing to increase the admissible angular acceleration or deceleration, for example for a boat from usual $40 \text{ rad}\cdot\text{s}^{-2}$ to more than $50 \text{ rad}\cdot\text{s}^{-2}$, and for example for a rolling mill from $100 \text{ rad}\cdot\text{s}^{-2}$ to more than $130 \text{ rad}\cdot\text{s}^{-2}$.

The first and second fastening means are less stressed improving the duration of the machine **4**.

In another variant of the magnetic masses **24**, **25** illustrated on FIG. 3, each magnetic mass **24**, **25** comprises two

compaction plates **28a** connected together with ties rods (not represented) and tightening the magnetic laminated sheets **27** to maintain the stacked magnetic laminated sheets compacted.

Each compaction plate **28a** comprises through holes **28b** so that conductive bars **31a** regularly arranged according to a diameter of the magnetic mass **24**, **25** are lodged in the housing **30** and go through the through holes **28b**.

Each end of the conductive bars **31a** is connected to one of the two short circuit rings **29a** located at each end of the magnetic mass **24**, **25** and exceeding the magnetic mass **24**, **25**.

As the short circuit rings **29a** exceed the magnetic mass **24**, **25**, the air gap **26a** between the two magnetic masses **24**, **25** is reduced compared to the air gap **26** separating the magnetic masses **24**, **25** as illustrated on FIG. 2.

The variant of the magnetic masses **24**, **25** illustrated on FIG. 2 permits to bring closer the magnetic masses **24**, **25** in the longitudinal direction compared to the variant of the magnetic masses **24**, **25** illustrated on FIG. 3 comprising short circuit rings **29a** exceeding the magnetic masses **24**, **25**.

Bringing closer the magnetic masses **24**, **25** in the longitudinal direction permits to reduce the length of the stator **21** so that the length of the magnetic mass **23** of the stator **21** is reduced reducing the weight of the stator **21**, and so that the length of the coils in the magnetic mass **23** is reduced. The longitudinal length of the machine **4** is reduced so that the machine **4** is more compact and less heavy.

When the machine **4** is an asynchronous machine, the first and second magnetic masses **24**, **25** comprise a squirrel cage or coils.

In case of a defect of the machine **4**, the supply of the coils of the stator **21** is stopped or no current is induced in the squirrel cage so that no magnetic field is generated preventing the propagation of the failure.

Similarly, when the machine **4** is a synchronous machine, the direct current supply of the coils inserted in the first and second magnetic masses **24**, **25** is stopped in case of a defect so that no magnetic field is generated preventing the propagation of the failure.

FIG. 4 illustrates a partial longitudinal cross section of a second embodiment of the moving set **100**.

The second embodiment of the moving set **100** differs from the first embodiment of the moving set **100** illustrated on FIG. 2 in that the diameter of the first shaft collar **34** referenced **D36** is smaller than the diameter of the second cylindrical surface **37a** of the second shaft collar **37** referenced **D37a**.

The first and second shoulders **36**, **37** are oriented on the first and second shaft collars **34**, **35** so that the second annular surface **37b** is facing the surface **40** of the first shoulder **36** opposite to the first annular surface **36b** of the first shoulder **36**.

The relative orientation of the first and second shoulders **36**, **37** one to the other on the first and second shaft collars **34**, **35** and the relative values of the diameters **D36** and **D37a** one to the other makes the mounting or dismounting of the first and second magnetic masses **24**, **25** easier on the shaft **5** when the first and second rotor is inserted in the stator **21**, for example during repairing or maintenance operations of the machine **4**.

For example, during the dismounting of the first and second magnetic masses **24**, **25**, the first fixing means are removed and the first magnetic mass **24** is removed from the stator **21** by translating the first magnetic mass **24** along the longitudinal direction.

Then the first magnetic mass **24** is removed from the stator **21**, the second fixing means are removed.

As the diameter **D36** is smaller than the diameter **D37a**, the second magnetic mass **25** is removed from the stator **21** by translating the second magnetic mass **25** along the longitudinal direction.

During the mounting of the first and second magnetic masses **24**, **25**, the second magnetic mass **25** is inserted in the stator **21** by translating the second magnetic mass **25** along the longitudinal direction until the second full rim comes in abutment into the second shoulder **37**.

The second fixing means are inserted to fix the second full rim on the second shaft collar **35**.

Then, the first magnetic mass **24** is inserted in the stator **21** by translating the first magnetic mass **24** along the longitudinal direction until the first full rim comes in abutment into the first shoulder **36**, and the first fixing means are inserted to fix the first full rim on the first shaft collar **34**.

In variant, the first and second magnetic masses **24**, **25** are disposed in the stator **21** and the shaft **5** is translated in the stator **21** along the longitudinal direction until the first full rim comes in abutment into the first shoulder **36**, and the first and second fixing means are inserted to fix the first full rim on the first shaft collar **34** and to fix the second full rim on the second shaft collar **35**.

FIG. 5 illustrates a partial longitudinal cross section of a third embodiment of the moving set **100**.

The third embodiment of the moving set **100** differs from the first embodiment of the moving set **100** in that the shaft **5** comprises two half shafts **5a**, **5b** and the second shaft collar **35** comprises two parts **35a**, **35b**.

The first half shaft **5a** comprises the first part **35a** and the first shaft collar **34**.

In variant, the second half shaft **5b** comprises the first shaft collar **34**.

The second half shaft **5b** comprises the second part **35b**.

The first part **35a** comprises the second annular surface **37a** and the second part **35b** comprises the second cylindrical surface **37a**.

The first and second half shafts **5a**, **5b** are fixed together with second removable fastening means to form the second shoulder **37** of the second shaft collar **35**.

The second removable fastening means may comprise bolts **41**.

In variant, the first part **35a** comprises the second cylindrical surface **37a** and the second part **35b** comprises the second annular surface **37a**.

In variant, the first shaft collar **34** is formed from the first and second parts.

The shaft **5** comprising the two half shafts **5a**, **5b** fixed together with the second removable fastening means is easily dismountable when the shaft **5** and one or both of the engine shaft **3** and the propeller shaft **12** are made of one shaft.

FIG. 6 illustrates a partial longitudinal cross section of a fourth embodiment of the moving set **100**.

The shaft **5** comprises a first half shaft **5c** and a second half shaft **5d**.

The moving set **100** further comprises a hollow cylindrical support part **42** inserted between the first half shaft **5c** and the second half shaft **5d**.

In this embodiment, the first and second fastening means comprise the hollow cylindrical support part **42**.

The cylindrical support part **42** is fixed on the first half shaft collar **5c** and the second half shaft collar **5d** with removable fastening means, for example screws **70** regularly

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disposed on a diameter of the said half shafts **5c**, **5d**, passing through the cylindrical support part **42**, and inserted into the cylindrical support part **42**.

The first and second magnetic masses **24**, **25** comprise the laminated magnetic sheets **27** compacted between the two compaction plates **28** and are fixed on the periphery of the cylindrical support part **42**.

The fixation of the first and second magnetic masses **24**, **25** on the half shafts **5c**, **5d** with the hollow cylindrical support part **42** permits to rigidify the fixation of the first and second magnetic masses **24**, **25** on the shaft **5**.

As the cylindrical support part **42** is fixed with removable fastening means on the first and on the second half shafts, the cylindrical support part **42** is fixed on the shaft **5** with two fixing interfaces, each interface comprising the surface of the cylindrical support part **42** in contact with a half shaft **5c**, **5d** and comprising the surface of the said half shaft **5c**, **5d** in contact with the cylindrical support part **4** so that the moving set **100** may endure higher decelerations or accelerations without damaging the moving set **100**.

Further, as the cylindrical support part **42** is hollowed, the inertia of the moving set **100** is not increased.

Each magnetic mass **24**, **25** may further comprise permanent magnets **71** so that the shaft **5**, the support part **42**, and the first and second magnetic masses **24**, **25** form two permanent magnet rotors, the machine **4** being a permanent magnet machine having two rotors.

In variant, the permanent magnets **71** may be replaced by a squirrel cage to form an asynchronous squirrel cage machine having two rotors.

In another variant, the permanent magnets **71** may be replaced by coils to form an asynchronous wound machine having two rotors or a synchronous wound machine having two rotors.

FIG. **7** illustrates a partial longitudinal cross section of a fifth embodiment of the moving set **100**.

The shaft **5** comprises a third full rim **43** and a fourth full rim **44** enclosing the shaft **5**.

The shaft **5**, the third full rim **43** and the fourth full rim **44** may be a single piece obtained for example by forging, casting, or welding the full rims **43**, **44** on the shaft **5**.

Each full rim of the third full rim **43** and a fourth full rim **44** comprises an annular support part **43a**, **44a** and an intermediary part **43b**, **44b** connecting the annular support part **43a**, **44a** to the shaft **5**.

The first and second magnetic masses **24**, **25** comprise the laminated magnetic sheets **27** compacted between the two compaction plates **28** and are fixed on the periphery of the annular support part **43a**, **44a** with the first and second fastening means comprising for example screws **45** passing through the support parts **43a**, **44a** and inserted in the magnetic masses **24**, **25**.

The fixation of the first and second magnetic masses **24**, **25** on the third full rim **43** and the fourth full rim **44** permits to rigidify the fixation of the first and second magnetic masses **24**, **25** on the shaft **5** so that the moving set **100** may endure higher decelerations or accelerations without damaging the moving set **100**.

Each magnetic mass **24**, **25** may further comprise the short circuit disks **29** and the conductive bars **31** lodged in the housing **30**, the first magnetic mass **24** comprising the first squirrel cage and second magnetic mass **25** comprising the second squirrel cage, the machine **4** being an asynchronous squirrel cage comprising two rotors.

In variant, the first and second squirrel cages may be replaced by permanent magnets to form a permanent magnet machine having two rotors.

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In another variant, the first and second squirrel cages may be replaced by coils to form an asynchronous wound machine having two rotors or a synchronous wound machine having two rotors.

FIGS. **8** and **9** illustrates a sixth embodiment of the moving set **100**.

The FIG. **8** illustrates a partial view of the sixth embodiment of the first magnetic mass **25** according to a longitudinal direction of the machine **4**, and FIG. **9** illustrates a partial longitudinal cross section of the sixth embodiment of the moving set **100**.

The first cylindrical magnetic mass **24** is segmented in first segments **24a**, **24b**, **24c**, **24d**, **24e**, **24f**.

Similarly, the second cylindrical magnetic mass **25** is segmented in second segments. For reasons of clarity, only a second segment **25a** is represented.

In variant, only one magnetic mass of the first and second magnetic masses **24**, **25** is segmented.

Each first segment **24**, **24b**, **24c**, **24d**, **24e**, **24f** comprises a first stack of segmented laminated magnetic sheets **27a** and a first support part **46** separating the first stack in two sets of laminated magnetic sheets **27**.

As represented, the first support part **46** may be in the middle of the first stack.

Similarly, each second segment **25a** comprises a second stack of segmented laminated magnetic sheets **27a** and a second support part **47** separating the second stack in two sets of laminated magnetic sheets **27**.

Each segment of the first segments **24a**, **24b**, **24c**, **24d**, **24e**, **24f** and second segments **25a** comprises two segments **28b** of segmented compaction plates so that the segmented laminated magnetic sheets **27** of each first and second segments is compacted between the two segments **28b** of segmented compaction plates.

Each segment of the first segments **24a**, **24b**, **24c**, **24d**, **24e**, **24f** and second segments **25a** further comprises two segments **29b** of segmented short circuit disks and the conductive bars **31** lodged in the housing **30** so that the first segments **24**, **24b**, **24c**, **24d**, **24e**, **24f** of the first magnetic mass **24** form the first squirrel cage and the second segments **25a** of the second magnetic mass **25** form the second squirrel cage. The machine **4** is an asynchronous squirrel cage comprising two rotors.

The rods **72** passing through the laminated magnetic sheets **27a**, the segment **28b** of the segmented compaction plates, the segment **29b** of the segmented short circuit disks, and the support parts **46**, **47** maintain the first and second stacks compacted between the segments **29b** of the short circuit disks **29** and the segments **28b** of the compaction plates.

The tie rods **72** are for example fixed with nuts **48** on the segments **28b** of the compaction plates **28**.

The shaft **5** may comprise the first shaft collar **34** comprising the first shoulder **36** and the second shaft collar **35** comprising the second shoulder **37**.

The first support part **46** is in abutment against the first shoulder **36** and is fixed on the first shaft collar **34** with the first fixing means comprising for example the screws **38**.

The second support part **47** is in abutment against the second shoulder **37** and is fixed on the second shaft collar **35** with the second fixing means comprising for example the screws **38**.

The splitting of the magnetic mass **24**, **25** in segments **24**, **24b**, **24c**, **24d**, **24e**, **24f**, **25a** permits to easily transport the moving set **100** in a plurality of parts and to facilitate the maintenance of the machine **4**.

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If the moving set **100** is defect, only the defect segment(s) **24**, **24b**, **24c**, **24d**, **24e**, **24f**, **25a** need to be dismantled and repaired or replaced.

In variant, the first and second squirrel cages may be replaced with permanent magnets to form a permanent magnet machine having two rotors.

In another variant, the first and second squirrel cages may be replaced by coils to form an asynchronous wounded machine having two rotors or a synchronous wounded machine having two rotors.

FIG. **10** illustrates a partial longitudinal cross section of a seventh embodiment of the moving set **100**.

The seventh embodiment of the moving set **100** differs from the first embodiment of the moving set **100** illustrated on FIG. **2** in that each conductive bar **31b** of the first set of conductive bars **31** of the first magnetic mass **24** is connected to a different conductive bar **31c** of the second set of conductive bars **31** of the second magnetic mass **25**, in that each magnetic mass **24**, **25** comprises a perforated compaction plate **28c** replacing a compaction plate **28** so that the conductive bars **31b**, **31c** may be joined together, and in that each magnetic mass **24**, **25** comprises only one short circuit disk **29**.

The bars **31b** of the first set of conductive bars **31**, the bars **31c** of the second set of conductive bars **31**, and the short circuit disks **29** form a squirrel cage so that the first and second cylindrical magnetic masses form a squirrel cage rotor, the machine **4** being an asynchronous machine having one rotor.

The number of short circuit disks **29** is divided by two compared to the first embodiment of the moving set **100** illustrated on FIG. **2** so that the weight of the moving set **100** is less than the weight of the first embodiment of the moving set **100**.

The end of each conductive bar **31b** of the first set in contact with a conductive bar **31c** of the second set and the end of each conductive bar **31c** of the second set in contact with a conductive bar **31b** of the first set may comprise a slanted side **49** to compensate an axial or radial offset between the conductive bar **31b** of the first set and the conductive bar **31c** of the second set.

The conductive bars **31b**, **31c** may be fixed together with bolts **50**.

In variant, the slanted sides **49** of the conductive bars **31b**, **31c** are welded or brazed.

The connection of the conductive bars **31a**, **31b** of the first set and the second set together permits to reduce the air gap **26a** between the first and second magnetic masses **24**, **25** according to the embodiment of magnetic masses **24**, **25** illustrated on FIG. **3** comprising the conductive bars **31a**, **31b** exceeding the compaction plates **29a** so that the moving set **100** is more compact and lighter.

The machine **4** comprising the magnetic masses **24**, **25** according to the embodiment of the magnetic masses **24**, **25** illustrated on FIG. **3** wherein the conductive bars **31a**, **31b** of the first set and the second set are connected together is lighter and more compact.

In variant, each pair of conductive bars **31a**, **31b** of the first set and the second set connected together is replaced by a unique bar so that the set of unique bars and the short circuit disks **29** of the first and second magnetic masses **24**, **25** form the squirrel cage.

When the magnetic masses **24**, **25** is made according to the embodiment of magnetic masses **24**, **25** illustrated on FIG. **3**, the two magnetic masses **24**, **25** are fixed on the shaft **5**.

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The unique bars are inserted in the housings **30** according to the longitudinal direction.

For each insertion of a unique bar, the shaft may be rotated so that the housings **30** are sequentially in line with a tooling inserting the unique bar in the housings **30**.

When the first magnetic mass **24** comprises a first set of coils and the second magnetic mass **25** comprises a second set of coils, one end of a coil of the first set of coils may be connected to one end of a coil of the second set of coils with for example a flexible junction forming a wounded rotor so that the machine **4** may be an asynchronous wounded machine or a synchronous wounded machine comprising one rotor.

In variant, each pair of coils of the first set of coils and the second set of coils connected together is replaced by a unique coil.

FIG. **11** illustrates another embodiment of the ends of the conductive bars of the first and second sets connected together.

Each end of the conductive bars **31c**, **31d** of the first and second sets comprises a shim **51** to compensate an angular offset between the said conductive bars **31a**, **31b**.

The conductive bars **31c**, **31d** may be fixed together with bolts **50**.

In variant, the shims **51** of the conductive bars **31b**, **31c** are welded or brazed.

FIG. **12** illustrates a partial longitudinal cross section of an eighth embodiment of the moving set **100**.

The eighth embodiment of the moving set **100** differs from the first embodiment of the moving set **100** illustrated on FIG. **2** in that the first squirrel cage of the first magnetic mass **24** and the second squirrel cage of the second magnetic mass **25** are respectively replaced by a first set of coils **80** and a second set of coils **81**.

The first set of coils **80** is regularly disposed according to a diameter of the first magnetic mass **24** and the second set of coils **81** is regularly disposed according to a diameter of the second magnetic mass **25**.

The first and second set of coils **80**, **81** are supplied respectively by first and second supply means **82**, **83** comprising for example wires.

The machine **4** is a synchronous machine comprising two wounded rotors, a first rotor comprising the first magnetic mass **24** and the second rotor comprising the second magnetic mass **25**.

In variant, the first and second set of coils **80**, **81** are supplied by a same supply means.

In another, each coil of the first of coils **80** is connected with a different coil of the second of coils **81** to form one rotor, the machine **4** being a synchronous machine comprising one wounded rotor.

Two or more of the embodiments of the movable set **100** represent on the FIGS. **2**, **3**, **4**, **5**, **7**, **8**, **9**, **10**, **11**, **12** may be combined.

FIG. **13** illustrates a partial longitudinal cross section of a ninth embodiment of the moving set **100**.

The shaft **5** is a single shaft comprising a shaft collar **90**.

The first and second magnetic masses **24**, **25** are fixed on the periphery of the shaft collar **90** respectively with the first and second fastening means comprising for example screws (not represented).

The first and second magnetic masses **24**, **25** comprise the laminated magnetic sheets **27** compacted between the two compaction plates **28** and are fixed on the periphery of the shaft collar **90**.

Each magnetic mass **24**, **25** may further comprise the permanent magnets **71** so that the shaft **5** and the first and

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second magnetic masses **24**, **25** form two permanent magnet rotors, the machine **4** being a permanent magnet machine having two rotors.

In variant, the permanent magnets **71** may be replaced by a squirrel cage to form an asynchronous squirrel cage machine having two rotors.

In another variant, the permanent magnets **71** may be replaced by coils to form an asynchronous wounded machine having two rotors or a synchronous wounded machine having two rotors.

In another embodiment (not illustrated), the rotating electrical machine **4** may be mounted in series with another rotating electrical machine as described above, the shaft **5** of the rotating electrical machine **4** being connected to the shaft of the other machine forming a set of rotating electrical machines.

FIG. **14** illustrates an example of a rolling mill **91** comprising two rolls **92** driven by the machine **4**.

A strand of steel **93** is inserted between the two rolls **92** to be laminated.

The invention claimed is:

1. A rotating electrical machine, comprising:

a stator;

a shaft centered in the stator; and

a first cylindrical magnetic mass and a second cylindrical magnetic mass, the first cylindrical magnetic mass and the second cylindrical magnetic mass enclosing the shaft and arranged in series on the shaft, the first cylindrical magnetic mass and the second cylindrical magnetic mass being separated by an air gap, the stator including coils, each coil being opposite to the two cylindrical magnetic masses, each cylindrical magnetic mass comprising a stack of compacted laminated magnetic sheets, a first fastener configured to fix the first cylindrical magnetic mass and the shaft, and a second fastener configured to fix the second cylindrical magnetic mass and the shaft, a diameter of the first cylindrical mass being equal to a diameter of the second cylindrical mass, the rotating electrical machine further comprising a hollow cylindrical support part, the shaft comprising a first half shaft and a second half shaft, the cylindrical support part being inserted between the first half shaft and the second half shaft, and the first cylindrical magnetic mass and the second cylindrical magnetic mass being fixed on the periphery of the cylindrical support part,

the first and second fasteners comprising the hollow cylindrical support part.

2. The rotating electrical machine according to claim 1, wherein each cylindrical magnetic mass comprises a set of coils so that the first cylindrical magnetic mass forms a first wounded rotor and the second cylindrical magnetic mass forms a second wounded rotor.

3. The rotating electrical machine according to claim 1, wherein each cylindrical magnetic mass comprises a set of coils, each coil of the set of coils of the first cylindrical magnetic mass being connected to a different coil of the set of coils of the second cylindrical magnetic mass so that the first and second cylindrical magnetic masses form a wounded rotor.

4. The rotating electrical machine according to claim 1, wherein each cylindrical magnetic mass comprises conductive bars and two short circuit disks or two short circuit rings to form a squirrel cage, the first cylindrical magnetic mass forming a first squirrel cage rotor and the second cylindrical magnetic mass forming a second squirrel cage rotor.

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5. The rotating electrical machine according to claim 1, wherein the first cylindrical magnetic mass comprises a first set of conductive bars and a first short circuit disk and the second cylindrical magnetic mass comprises a second set of conductive bars and a second short circuit disk, each conductive bar of the first set being connected to the first short circuit disk and to a different conductive bar of the second set, the conductive bars of the second set being connected to the second short circuit disk, the first and second sets and the first and second short circuit disks forming a squirrel cage so that the first and second cylindrical magnetic masses form a squirrel cage rotor.

6. The rotating electrical machine according to claim 5, wherein the end of each conductive bar of the first set and the second set in contact with another conductive bar comprises a slanted side.

7. The rotating electrical machine according to claim 1, further comprising a third full rim and a fourth full rim enclosing the shaft, the first cylindrical magnetic mass being fixed on the periphery of the third full rim with the first fastener, and the second cylindrical magnetic mass being fixed on the periphery of the fourth full rim with the second fastener.

8. The rotating electrical machine according to claim 1, wherein at least one magnetic mass of the first cylindrical magnetic mass and the second cylindrical magnetic mass is segmented in segments, each segment comprising a stack of segmented laminated magnetic sheets and a segmented support part separating the stack in two sets of segmented laminated magnetic sheets, a respective first or second fastener associated with one magnetic mass fixing each support part on the shaft.

9. A set of rotating electrical machines comprising at least a first rotating electrical machine and a second rotating electrical machine according to claim 1, the shaft of the first rotating electrical machine being connected to the shaft of the second rotating electrical machine.

10. A boat comprising at least one rotating electrical machine according to claim 1.

11. A rolling mill comprising at least one rotating electrical machine according to claim 1.

12. A rotating electrical machine, comprising:

a stator;

a shaft centered in the stator; and

a first cylindrical magnetic mass and a second cylindrical magnetic mass, the first cylindrical magnetic mass and the second cylindrical magnetic mass enclosing the shaft and arranged in series on the shaft, the first cylindrical magnetic mass and the second cylindrical magnetic mass being separated by an air gap, the stator including coils, each coil being opposite to the two cylindrical magnetic masses, each cylindrical magnetic mass comprising a stack of compacted laminated magnetic sheets, a first fastener configured to fix the first cylindrical magnetic mass and the shaft, and a second fastener configured to fix the second cylindrical magnetic mass and the shaft;

a first full rim including a first central circular aperture, the first magnetic mass being fastened on the periphery of the first full rim;

a second full rim including a second central circular aperture, the second magnetic mass being fastened on the periphery of the second full rim;

a first shaft collar and a second shaft collar on the shaft, wherein:

the first fastener comprises a first shoulder on the periphery of the first shaft collar and first fixing means;

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the first full rim is into abutment against the first shoulder and is fixed to the first shaft collar by the first fixing means;

the second full rim is into abutment against the second shoulder and is fixed to the second shaft collar by a second fixing means;

the first shoulder comprises a first cylindrical surface in contact with the circumferential surface of the first central circular aperture and a first annular surface perpendicular to the first cylindrical surface in contact with the first full rim; and

a second shoulder comprises a second cylindrical surface in contact with the circumferential surface of the second central circular aperture and a second annular surface perpendicular to the second cylindrical surface and in contact with the second rim.

13. The rotating electrical machine according to claim **12**, wherein the diameter of the first shaft collar is smaller than the diameter of the second cylindrical surface, and the first

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and second shoulders are oriented on the first and second shaft collars so that the second annular surface is facing the surface of the first shoulder opposite to the first annular surface of the first shoulder.

14. The rotating electrical machine according to claim **12**, wherein the shaft comprises two half shafts, and wherein one shaft collar of the first and second shaft collars comprises two parts, a first half shaft comprising a first part and the second half shaft comprising the second part, the first part comprising the annular surface or the cylindrical surface of the shoulder of the said one shaft collar and the second part comprising the cylindrical surface or the annular surface of the shoulder of the said one shaft collar, the first and second half shafts being removably fixed together with the second fastener to form the shoulder of the said shaft collar, the first half shaft or the second half shaft comprising the other shaft collar of the first and second shaft collars.

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